

Assignment - 1

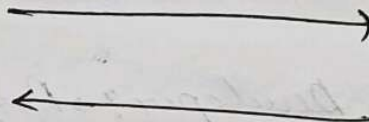
Question 1:

Aspect	Frontend	Backend	Full-stack.
Definition	Builds the user interface (UI).	Handles server, database and business logic.	Develops both frontend and backend.
Focus	What users see and interact with.	Data processing, authentication, APIs.	Complete web application development.
Technologies	HTML, CSS, JS, React, Bootstrap.	Node.js, Express, Python, Java, PHP, MongoDB.	MERN, LAMP, Django + React.
Example	Amazon search bar and product cards.	Amazon payment processing and database updates.	Developing a complete food delivery website.

Question 2:

CLIENT
(web Browser/User)

1. HTTP Request (Get / Index.html)



Web Server
(Host / Application)

2. HTTP Response (HTML, CSS, JS)



Question 3:

1. URL Input: User enters a web address (e.g. <https://example.com>) into the browser.

2. DNS Lookup: Browser contacts a DNS server to resolve domain name into an IP address (e.g. 93.184.216.34).

3. HTTP/HTTPS Request: Browser sends a Get request over TCP/IP connection to the server.

4. Server Processing: The web server receives request, fetches assets / data.

5. HTTP Response.

6. DOM Tree Parsing

7. Render Tree & Painting

Question 1: Code Editor (VS Code): Text editor to write, edit, and debug HTML, CSS and JS code efficiently with extensions.

Web Browser & Dev Tools: To preview web pages and inspect DOM elements, network calls, and console logs.

Version Control System (Git & GitHub): Tracks code changes management, and remote code collaboration.

Terminal / Command Line Interface (Git Bash):

Executes scripts, manages file directories, and runs Git versioning commands.

Node.js & npm: JavaScript runtime environment and package manager to run build tools, servers, and dependencies.

Question 5: A web server refers to hardware or software that stores web asset files (HTML, CSS, JS, images) and serves them to clients across the internet using HTTP/HTTPS protocols.

- Apache HTTP server.

- NGINX

- Apache Tomcat

- Express.js.

Question 6: • Frontend Developer: → Develops the user interface.

- HTML, CSS, JS, React, Vue, Bootstrap

- Backend Developer: → Manages servers, database and logic.

- Node.js, Express, Python, Java, SQL, MongoDB

- Amazon payment processing authentication, database updates.

- Fullstack Developer: → Develops both frontend and backend.

- MEAN, LAMP, Django + React

- Building a complete food delivery website/app.

- Question 7:
- Installed VS code from the official website.
 - Installed extensions:
 - Live Server.
 - Prettier
 - Auto Rename Tag
 - ES7 + React / Redux Snippets.
 - Enabled editor • Format on Save: found in settings. json.
 - Sources Shot: Attach your VS Code Setup.

<u>Question 8:</u> Static Website	Dynamic website
→ Fixed HTML/CSS pages.	→ Generated dynamically from server / database
→ Same content for all users.	→ Personalized Content for each user
→ Portfolio or company website.	→ Amazon, Youtube, Twitter.

<u>Question 9:</u> Browser	Rendering Engine	Key Features.
Google Chrome.	Blink	Fast, modern web support.
Mozilla Firefox	Gecko	Privacy - focused.
Apple Safari	WebKit	Optimized for macOS / iOS.
Microsoft Edge	Blink	Chromium
Opera.	Blink	Compatibility Built-in ad & feature blocking.

Question 10:

